Σ^0 Production in the NuMI Beam at MicroBooNE: Analysis Note

J. Nowak, N. Patel

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Abstract

Working note for the MicroBooNE Σ^0 production analysis in the NuMI beam. The selection reconstructs the $\Sigma^0 \to \Lambda\gamma$, $\Lambda \to p\pi^-$ chain in $\bar{\nu}_\mu$ charged-current interactions and is finished with a cross-validated gradient BDT. This note records the current state, the planned port into the `xsec_analyzer` framework, and the three pieces of work that follow: a Poisson-thrown fake-data sample, extension from RHC-only to the combined FHC+RHC exposure, and the selection work needed to lift a purity that currently sits near 0.1%.

Contents

1 Status and scope

The analysis measures, or more realistically sets a limit on, Σ^0 production in $\bar{\nu}_\mu$ charged-current interactions. The signature is

$$\bar{\nu}_\mu + \mathrm{Ar} \to \mu^+ + \Sigma^0 + X, \qquad \Sigma^0 \to \Lambda\gamma, \qquad \Lambda \to p\,\pi^-, \tag{1}$$

so the reconstructed final state is a muon, a proton, a pion and a photon.

Table 1: Current selection performance, RHC only.

Stage	Raw signal	Efficiency [%]	Purity [%]	$S/\sqrt{S+B}$
Preselection (NTracksCut)	650	15.7	0.040	0.038
Cut-based	497	12.0	0.077	0.045
BDTG (current final)	502	12.1	0.108	0.054

The existing selection (`Sigma/pelee/SigmaSelection.C`) runs on standard PeLEE ntuples and reaches, at the operating point BDTG > -0.90 scaled to the full RHC data exposure of 1.1526×10^{21} POT:
Efficiency is respectable; purity is not. A purity of 0.108% means roughly one signal event per thousand selected, and ~ 2640 background events under ~ 2.9 signal events at full RHC POT. **Improving the selection is therefore the priority**, ahead of any normalisation or extraction work: at this purity no downstream treatment can rescue the measurement.

What this analysis produces. Not a differential cross section. At 0.1% purity and 12% efficiency the deliverable is a cross-section *upper limit*. This matters for the framework port (Sec. 2), because `xsec_analyzer` is built end-to-end for differential extraction and contains no limit machinery.

2 Framework port

The selection is being ported into `xsec_analyzer` as a `SelectionBase` subclass, alongside the CC1μ1π selections. The port buys the framework's systematic machinery – flux (PPFX multisims), cross-section (GENIE multisims and unisims), detector variations, hadron reinteraction – which is exactly the covariance a limit needs for its background prediction, and which would otherwise have to be built from scratch.

What is ported. Selection, cut flow, truth categorisation, reco and true observables, and the systematic universes at a *single* analysis bin.

What is not. Limit setting, which stays in `CrossSection.C` reading the framework covariance. An unfolded Σ^0 spectrum would be meaningless at this purity, so no multi-bin scheme is planned.

Framework gaps. Three, all small. Nine branches the selection uses are present in the ntuples but unbound in `Branches.hh` (`trk_pid_chipi_v`, `shr_energy_y_v`, `shr_p{x,y,z}_v`, `pi0_energy{1,2}_Y`, `pi0_gammadot`, `shr_moliere_avg_v`); the fiducial volume is looser than the CCπ one and must be set per selection rather than inherited; and the factory needs a registration branch. Details in `SIGMA0_FRAMEWORK_PLAN.md`.

3 Selection

3.1 Signal definition

`nu_pdg` $= -14$, `ccnc` $= 0$, a Σ^0 (PDG 3212) in the final-state list `mc_pdg`, and the true vertex inside the fiducial volume. Photons are not stored in `mc_pdg`, so the decay is tagged by the Σ^0 itself rather than by its daughters.

3.2 Fiducial volume

$3.00 < x < 253.35$, $-112.53 < y < 114.47$, $3.1 < z < 1033.0$ cm, excluding the dead region $675.1 < z < 775.1$ cm. This is deliberately looser than the CC1μ1π box (10–246, ±101, 10–986): the signal is rare enough that acceptance is worth more than vertex quality.

3.3 Candidate reconstruction

- **Muon**: longest track with `trk_score` > 0.8, start within 20 cm of the vertex, length > 10 cm, and LLR PID > 0.7.
- **Proton and pion**: assigned jointly by the combined discriminant

$$D = \frac{\mathrm{LLR}}{0.591} + \frac{\chi_p^2 - \chi_\pi^2}{124.5}, \tag{2}$$

  with proton $= \arg\min D$ and pion $= \arg\max D$. This separates true protons from true pions with AUC 0.873, against 0.838 for χ_p^2 alone and 0.852 for LLR alone. The χ^2 term is dropped per track when undefined ($\sim$30% of tracks).
- **Photon**: the two non-track PFPs closest to the vertex; the closest is taken as the Σ^0 decay photon.

3.4 Cut chain

Nine cumulative stages: `AllEvents`, `FiducialV`, `CosmicVeto` (`topological_score` > 0.20), `NShowers` ($0 < n_{\mathrm{shr}} < 4$), `MuonCandidate`, `ProtonCandidate`, `PionCandidate`, `NTracksCut` ($n_{\mathrm{trk}} < 5$), `BDTG`.

3.5 Derived observables

Λ mass from (p, π); Σ^0 mass from (Λ, γ) using the per-PFP shower energy `shr_energy_y_v`; Λ and photon vertex-pointing angles and impact parameters; Λ decay distance; and a π^0 di-photon mass built from the framework π^0 reconstruction for background rejection.

3.6 BDT

Gradient BDT over 49 inputs, evaluated with two-fold cross-validation: folds are trained on even and odd event numbers and each is applied only to the events it did not train on. Per-fold ROC $\simeq 0.81$. The cross-validation is not optional – applying a fold to its own training events would inflate the ROC and the apparent significance.

4 Why the purity is low, and what to do about it

This is the priority. Two diagnostics point at the same place.

The surviving background is π^0 events. RES and DIS together are 76%. Both produce $\pi^0 \to \gamma\gamma$, which supplies a fake Σ^0 photon, plus charged hadrons that fake the proton and pion tracks. The background is not diffuse: it is one physics process.

The candidate assignment is the weak link. Truth-matched purities of the individual candidates in background events:

Table 2: Surviving background by interaction mode at the final selection.

Mode	Scaled yield	% of background
Resonant (RES)	1381.8	52.3
Deep inelastic (DIS)	635.3	24.0
Quasi-elastic (QE)	363.8	13.8
Meson exchange (MEC)	177.1	6.7
Cosmic	74.6	2.8
Coherent (COH)	8.9	0.3
Total	2643.1	100

Candidate	Correct [%]	Dominant impostor
Muon	81	—
Proton	58	—
Pion	31	another proton (28%)
Photon	29	a proton (18%)

The muon is reliable and the proton is acceptable, but **the pion and photon candidates are wrong roughly** 70% **of the time**. Since both the Λ and Σ^0 masses are built from them, the two most powerful BDT inputs are being computed from misidentified objects in most background events – and, importantly, we do not know how often in signal events, because the NoCR signal sample's non-muon truth matching is unreliable.

4.1 The photon candidate: what the data says

This was investigated first, and the result contradicts the obvious expectation, so it is recorded in full.

Composition. Truth-matching the photon candidate in background events (69557 preselected events) gives:

Truth match	Fraction [%]
photon	30.0
unmatched (PDG 0)	26.7
proton	15.2
charged pion	16.3
muon	9.8
electron	1.4

Charged tracks reconstructed as showers – proton, pion, muon together – account for 41%, and a further 27% has no truth match at all.

The signal photon is soft, which inverts the strategy. The Σ^0–Λ mass splitting is only 77 MeV, so

$$E_\gamma = \frac{m_{\Sigma^0}^2 - m_\Lambda^2}{2m_{\Sigma^0}} = 74.5 \text{ MeV} \tag{3}$$

in the Σ^0 rest frame. The measured mean energy of the photon candidate in signal events is 70 MeV, which matches that prediction and is good evidence that the selection is finding the right object. The background photon candidate averages 140 MeV, because the π^0 photons that dominate it are far harder.

Requiring shower-like quality would therefore remove signal preferentially, which is the opposite of what an earlier draft of this note proposed. A 74 MeV photon is near the reconstruction threshold: it makes few hits and deposits little energy. Measured against true photons in background events – which are π^0 photons and much harder – the signal photon looks *less* shower-like, not more.

Cutting in the correct direction helps, but only before the BDT. Scanning cuts in the soft direction at preselection:

Cut	ϵ_S [%]	Background rejected [%]	FoM gain
$E_\gamma < 120$ MeV	94.5	25.4	$\times 1.092$
$N^Y_{\text{hits}} < 102$	95.7	21.5	$\times 1.079$
$E_\gamma \in [5, 100]$ MeV	70.6	55.2	$\times 1.052$

But `Shower1Energy` is already one of the 49 BDT inputs, so the question is whether the cut adds anything the BDT has not already taken. Applied on top of the BDT operating point it does not:

Selection	S	B	FoM
BDT only	502	20762	3.443
BDT + $E_\gamma < 120$ MeV	482	18663	3.484 ($\times 1.012$)
BDT + $E_\gamma < 100$ MeV	456	17706	3.384 ($\times 0.983$)

Conclusion. The photon candidate cannot be improved by cutting on variables the BDT already has – it is using them close to optimally. An explicit π^0 mass veto is likewise ineffective, rejecting only 9% of background, because it requires *two* reconstructed showers and most background events supply only one. Improvement must come from information the BDT does not yet have.

The one genuinely unused handle. `shr_moliere_avg_v`, the shower Molière angle, is in the ntuples and in the plotting variable list but is *not* among the 49 BDT inputs. It measures lateral spread, which is orthogonal to energy and hit count and is the classic discriminant between an electromagnetic shower and a track stub. Adding it requires re-running the selection to dump it into the training tree, then retraining both folds. That is the concrete next step for the photon candidate, and it should be judged by the FoM gain on top of the BDT, not at preselection.

4.2 Proposed selection work, in order

1. **Add `shr_moliere_avg_v` to the BDT** (Sec. 4.1) and retrain. This is the only unused photon handle; cuts on the existing ones are exhausted.

2. **Improve the pion candidate (31% correct).** The joint discriminant of Eq. 2 is already a real improvement over χ^2_p alone, but the failure mode is specific: a second proton is taken as the pion 28% of the time. Adding the sign-sensitive information – track length against the proton range curve, or the Λ-mass constraint itself as part of the assignment rather than as a consequence of it – would target that directly.

3. **Reconsider the Λ decay topology.** Λ has $c\tau = 7.89$ cm, so a genuine $\Lambda \to p\pi^-$ has a displaced vertex. The selection computes `LambdaDecayDist` and the pointing angle but does not cut on them. A displaced-vertex requirement is close to background-free in principle and is the most distinctive feature of the signal.

The analysis is also **signal-statistics limited**: 650 raw signal events at preselection, with a BDT ROC that plateaus regardless of background size or added variables. More signal MC is the largest single lever, and no amount of selection work substitutes for it.

5 Fake data by Poisson throw

The problem. The current "data" sample is the signal file itself (`SigmaSelection.C` maps the same NuWro hyperon ntuple to both sample 0 and sample 4), scaled by POT. That is not a closure test: it contains no background, no statistical fluctuation, and is perfectly correlated with the signal prediction it would be compared against.

The fix. Adopt the same Poisson-throw scheme the CC$1\mu1\pi$ analysis uses (`macros/throw_perrun_*.C`). For each MC event with central-value weight $w_{\rm cv}$ and POT scale $\alpha = \mathrm{POT}_{\rm data}/\mathrm{POT}_{\rm MC}$, draw

$$n \sim \mathrm{Poisson}(w_{\rm cv}\,\alpha) \tag{4}$$

and write the event n times into the fake-data tree with all weights reset to unity. Applied across *all* samples – signal, other hyperons, background, dirt – this produces a fake dataset that

- contains the correct background admixture, which matters enormously here because background outnumbers signal by $\sim$900:1;
- carries the right statistical fluctuations, so uncertainties can be validated rather than assumed;
- is statistically independent of the prediction, making a genuine closure test possible.

Implementation. A `throw_sigma0.C` modelled on `throw_perrun_w.C`: chain the processed MC, drop the multisim weight branches, `CloneTree(0)` to preserve the Σ^0 observables, throw, and stamp `summed_pot` with the target POT. One seed per sample so the throw is reproducible.

Caveat to keep in view. A Poisson throw from the same MC is a *statistical* closure test, not a model test. It cannot reveal generator mismodelling, because the "data" and the prediction come from the same generator. A genuine model test needs a hyperon sample from a different generator – GENIE against the current NuWro – which does not exist locally. Until it does, the throw validates the statistical and normalisation chain only, and that limitation should be stated wherever the closure is quoted.

6 Exposure: RHC now, combined later

Now: RHC only. Everything is scaled to the full RHC data exposure, $\mathrm{POT}_{\rm data} = 1.1526\times10^{21}$:

Sample	Source	Generated POT
Σ^0 signal / hyperons	NuWro hyperon-enhanced	2.1764×10^{23}
Background	RHC overlay, Runs 1+2+4a+4b+4c	9.94482×10^{21}
Dirt	Run 1 RHC only	4.21274×10^{20}

Two known imperfections: the dirt sample is Run-1 RHC only, scaled up to the full RHC exposure, and the signal sample is generated with no cosmic overlay (NoCR), which is why its non-muon truth matching cannot be trusted.

Later: combined FHC+RHC. The signal is $\bar{\nu}_\mu$ charged-current, so RHC is the sensitive mode – but FHC carries a substantial wrong-sign $\bar{\nu}_\mu$ component (34.0% of the FHC flux, against 53.3% for RHC), and the FHC exposure is 8.857×10^{20} POT. Folding FHC in therefore adds real signal rather than only background, and the CC$1\mu1\pi$ analysis already has the machinery: per-run POT scaling, a combined `file_properties`, and combined flux normalisation.

Deferred deliberately. The combination is a normalisation exercise, and at 0.1% purity it would add background faster than signal in absolute terms. The selection work of Sec. 4 comes first; the exposure extension is worth doing once the selection is worth extending. When it happens it needs: FHC overlay samples processed through the same selection, an FHC hyperon signal sample (or a justified reuse of the RHC one, given the signal is beam-mode independent at the interaction level), FHC dirt, and a combined-mode flux normalisation following `Flux` in the CCπ xsec configs.

7 Work plan

1. **Selection (priority).** Photon-candidate identification, explicit π^0 veto, pion assignment, Λ displaced-vertex requirement. Sec. 4.
2. **Framework port.** Branch bindings, `Sigma0` selection class, factory registration, cut-flow validation against the standalone code before any config work. Sec. 2.
3. **Fake data.** `throw_sigma0.C`, all samples, RHC POT. Sec. 5.
4. **Systematics.** Single-bin `univmake` for the background covariance.
5. **Combined exposure.** FHC+RHC once the selection justifies it. Sec. 6.
6. **Limit.** Outside the framework, reading its covariance.